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Fairness in Pedestrian Detection: A Comparative Study of Regional and Demographic Bias Using YOLOX

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Declaration

I hereby certify that the material, which I now submit for assessment on the programmes of study leading to the award of Master of Science, is entirely my own work and has not been taken from the work of others except to the extent that such work has been cited and acknowledged within the text of my work. No portion of the work contained in this thesis has been submitted in support of an application for another degree or qualification to this or any other institution.



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Abstract

Autonomous driving systems promise transformative transportation solutions but face challenges of fairness and reliability when deployed across diverse contexts. This study investigates regional and demographic disparities in pedestrian detection using the YOLOX model, comparing performance across European and Indian street scenes. Significant biases are identified, including higher miss rates for Indian pedestrians (28.27%) versus European pedestrians (6.56%), gender-based unfairness in Indian datasets, and age-based bias in European datasets. A key achievement of this research is the creation of a custom Indian pedestrian dataset, comprising 2,173 images with 5,416 manually annotated bounding boxes across sensitive groups (male, female, children), tailored to the unique challenges of Indian traffic conditions. While these findings are specific to YOLOX and not generalizable to all pedestrian detectors, they emphasize the need for fairness checks in developing such systems. By integrating human-centric principles like equity, safety, and transparency, this research offers actionable insights into fairness testing, dataset augmentation, and model design, paving the way for safer and more inclusive autonomous driving solutions.

Keywords

Autonomous driving, pedestrian detection,fairness , YOLOX.

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1 Introduction

1.1 Introduction

Autonomous driving systems are revolutionizing transportation, offering enhanced safety and efficiency. At their core, pedestrian detection technologies have seen significant advancements, enabling applications in autonomous vehicles and urban surveillance. However, concerns about fairness and reliability arise as these systems are deployed across diverse regional and cultural contexts. Additionally, their susceptibility to software flaws [1] poses serious risks. The tragic 2018 incident involving an autonomous Uber vehicle, which resulted in a pedestrian fatality [2], highlights the potential dangers of such flaws. This underscores the urgent need for robust, equitable, and reliable detection systems to address both technical vulnerabilities and fairness issues, ensuring safe and inclusive transportation solutions for all.

Substantial research has been conducted on testing autonomous driving systems to enhance their reliability and robustness. For instance, Tian et al.[3] presented DeepTest, a technique leveraging image transformations to replicate camera noise typically encountered in autonomous driving environments. Similarly, Zhang et al.[4] proposed DeepRoad, utilizing Generative Adversarial Networks (GANs) to create synthetic test images based on real-world driving scenarios. Another notable contribution is DeepBillboard, introduced by Zhou et al.[5], which focuses on generating adversarial billboards designed to provoke steering errors in autonomous vehicles, thereby exposing potential system vulnerabilities. While numerous testing approaches have been developed in the autonomous driving industry, fairness testing remains an underexplored area. As an emerging field within software testing, fairness testing aims to identify and address biases in software systems[6], including AI-driven applications [7]. In the context of autonomous driving, fairness issues—such as higher detection accuracy for male pedestrians compared to females—can lead to discriminatory outcomes and unequal treatment. Such biases risk marginalizing certain groups, exacerbating existing social inequalities, and undermining trust in these systems. Addressing these challenges is essential to ensure that autonomous driving systems operate equitably and inclusively, making fairness testing a critical priority in their development and evaluation.

Examining fairness issues in autonomous driving systems across different geographic regions is crucial because traffic patterns, cultural contexts, and demographic characteristics vary significantly worldwide. India, for instance, presents a unique and challenging environment for testing such systems due to its complex traffic scenarios, diverse road user behaviors, and high pedestrian density. Unlike structured traffic systems often found in Western countries, Indian roads feature a mix of vehicles, pedestrians, animals, and non-standardized road infrastructure, all of which demand robust and adaptable detection capabilities.

Moreover, there is a significant possibility that widely used object detection and pedestrian detection models, such as YOLO or other pre-trained systems, have been trained predominantly on datasets originating from regions with well-organized traffic

and homogeneous demographic distributions[8]. These datasets may lack sufficient representation of Indian conditions, leading to unbalanced model training and potential fairness issues. For example, these models might perform well in detecting pedestrians in European or American street scenes but struggle to maintain similar accuracy in identifying Indian pedestrians or addressing diverse demographic groups, such as children or individuals wearing culturally specific attire.

1.1.1 Motivation

The motivation for this research stems from several critical observations:

Safety Implications: Biased pedestrian detection systems could pose serious safety risks when deployed in regions different from their training environments.

Global Deployment Challenges: As AI systems become more widespread, their ability to perform consistently across different geographical and cultural contexts becomes increasingly important.

Equity in AI Systems: There is a growing need to ensure AI systems serve all populations equally, regardless of their geographical or cultural context.

Limited Research: While bias in AI systems has been widely studied, regional bias in pedestrian detection systems remains understudied, particularly in comparing European and Indian contexts.

1.2 Problem Description

1.2.1 Research Questions and Objectives

The research is guided by two primary research questions:

RQ1:(Overall Fairness Comparison)

How do pre-trained object detection model YOLOX perform in pedestrian detection tasks across datasets from European and Indian street conditions?

Objectives:

- Evaluate and compare model performance across regional datasets.
- Quantify performance disparities.

RQ2:(Categry wise fairness) *To what extent the pre-trained pedestrian detectors exhibit unfairness concerning common sensitive groups?*

Objectives:

- Evaluate the fairness issue exhibited by models towards Indian and European dataset by analyzing the results across three sensitive groups - Male, Female, Children.

1.2.2 Relevance to Human-Centric AI

This research directly addresses key principles of Human-Centric AI:

Fairness and Inclusivity:

Investigating systemic biases in AI systems Ensuring equitable performance across different populations Promoting inclusive AI development practices

Safety and Reliability:

Identifying potential safety risks in deployment Improving model robustness across diverse environments Enhancing reliability in critical applications

Transparency and Trust:

Utilizing visualization techniques to understand model decisions Providing interpretable insights into bias sources Building trust through demonstrated fairness

2 Literature Review

Pedestrian detection is a key component of autonomous driving, involving the identification of pedestrians within street-level images through bounding boxes and confidence scores .Despite its importance, research addressing fairness in pedestrian detection systems remains limited. Early studies by Brandao[9] examined fairness using classic machine learning methods reliant on manually defined features. However, these methods have been superseded by modern deep learning-based techniques, which are now the norm in autonomous systems. Wilson et al.[10] investigated biases related to skin tone using a single dataset and two general object detectors, while Kogure et al.[11] analyzed age bias using a small-scale dataset with outdated detection methods. Collectively, these studies underscored disparities in detection accuracy, with darker-skinned individuals and children often being at a higher risk of non-detection.

A recent landmark study by Li et al,[12] delved into fairness issues in pedestrian detection systems, focusing on demographic disparities. This research revealed significant fairness concerns, particularly regarding lower detection rates for children and darker-skinned individuals. However, the scope of this study was limited to datasets representing European conditions, leaving questions about fairness in other regional contexts unanswered.

Research on pedestrian detection in Indian conditions is even scarcer. Nisha et al[13] examines the unique challenges posed by Indian road environments, such as varying lighting conditions, unstructured traffic patterns, and dense pedestrian activity. While this study highlights the need for robust detection systems in Indian contexts, it does not specifically address fairness issues or evaluate biases across sensitive groups such as males, females, and children.

Given this gap, our research aims to evaluate fairness in pedestrian detection systems within Indian conditions, specifically analyzing performance disparities across sensitive groups. By comparing these findings with fairness metrics established in European contexts, we seek to identify any notable discrepancies in detection fairness. This comparative analysis will provide insights into the global applicability and fairness of widely used object detection models, such as YOLO and other pedestrian detectors, and highlight potential areas for improvement to ensure equitable performance in diverse real-world environments.

3 Data Collection and Preparation

Data collection and preparation form the cornerstone of this research, ensuring that the comparative analysis of pedestrian detection performance across European and Indian street scenes is both robust and reliable. Properly curated datasets are critical to understanding and quantifying the regional biases inherent in CNN-based pedestrian detection systems. By employing representative datasets that accurately reflect the demographic and environmental characteristics of the two regions, this research aims to highlight disparities and propose actionable insights to address potential fairness issues in pedestrian detection algorithms. Given the objectives of this study, it is essential to utilize two types of datasets: one representing European street conditions and another representing Indian street conditions. The inclusion of these datasets is vital to capturing the diversity in pedestrian appearances, clothing styles, environmental lighting, infrastructure, and traffic patterns unique to each region. These distinctions are hypothesized to play a significant role in influencing model performance and highlighting biases that may arise from training datasets predominantly representing one region over another. This section outlines the overarching methodology for selecting and preparing datasets suitable for the research objectives. Detailed explanations of the data collection and preparation processes for both European and Indian datasets are provided in the subsequent subsections, where specific challenges, strategies, and quality assurance measures are discussed.

3.1 European Condition Datasets

The European condition datasets selected for this study include two notable publicly available datasets: the CityPersons dataset and the Berkeley Driving dataset (BDD100k). These datasets are widely recognized benchmarks for evaluating pedestrian detection systems and provide diverse environmental conditions and object annotations.

3.1.1 CityPersons Dataset

The CityPersons dataset stands as the most widely studied benchmark for evaluating pedestrian detectors. Its test set includes 1,525 images captured across six cities, showcasing diverse weather conditions and street scenes. The dataset includes various object labels such as Pedestrian, Rider, and Miscellaneous objects, making it a rich resource for evaluating pedestrian detection performance.

3.1.2 BDD100k Dataset

The BDD100k dataset is an extensive driving dataset comprising 2,233 images spanning 40 object classes typical of urban driving scenes. Captured across four different cities at various times of the day, the dataset features labels for a wide range of objects, including Person (Pedestrian/Rider), Vehicle (Car, Truck, Bus), and Traffic Elements,

among others. This diversity makes it an invaluable dataset for testing fairness and robustness in pedestrian detection systems.

3.2 Data Preparation for European Datasets

The preparation of the European datasets involved two key stages: extraction and filtering of relevant data for fairness analysis in RQ1 and manual annotation for sensitive subgroup analysis in RQ2. The final dataset used in this study is a combination of images from the CityPersons and BDD100k datasets, providing a diverse and comprehensive representation of European street scenes.

3.2.1 RQ1 Data Preparation

For fairness analysis under RQ1, the datasets were filtered to include only images containing the class of interest, "person." This ensured a focused evaluation on pedestrian detection. The widely adopted "reasonable subset" principle was followed to filter bounding boxes, selecting only those images with labeled pedestrians having a height of at least 50 pixels. This step minimized noise and ensured consistency in the analysis.

3.2.2 RQ2 Data Preparation

To address the fairness analysis in RQ2, manual annotations were added to label subcategories of interest (male/female/child) for each pedestrian in the dataset. The following methodology was employed for annotation:

1. Classification Process
    - Each pedestrian bounding box was manually reviewed to determine its subcategory.
    - If the individual visually appeared to belong to the "child" subcategory (infants, toddlers, or early school age), it was labeled accordingly. No further gender classification was performed for children.
    - For individuals not classified as children, gender was marked as either "male" or "female."
    - Bounding boxes where the subcategory could not be determined were ignored, and no additional subcategory label was added
2. Label File Format
    - Each bounding box included a "class" label as "person" and an optional "subcategory" label ("male," "female," or "child") based on the criteria above.
    - The annotations were stored in JSON format, with the file name matching the corresponding image file for easy reference
3. Quality Assurance
    - Each annotation was cross-verified to ensure accuracy and consistency.

An example JSON label file showcasing the subcategory annotations is provided in figure 3.1. This comprehensive labeling process enabled the datasets to support nuanced fairness analysis while maintaining high data integrity.

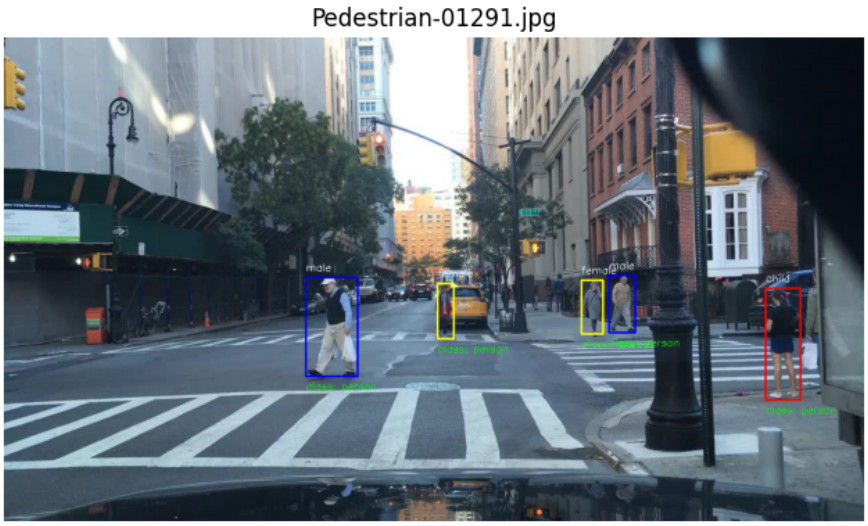


Figure 3.1: Pedestrians labeled with subcategory(from European Dataset).

3.2.3 Summary Statistics for the European Dataset

The final European dataset comprises 3,129 images in total, drawn from both the CityPersons and BDD100k datasets. The dataset includes the following bounding box statistics:

1. Total Number of Bounding Boxes (Persons): 9,082
2. Bounding Boxes by Subcategory:
    - Child: 423
    - Male: 5,063
    - Female: 3,596

3.3 Indian Condition Datasets

The Indian condition datasets used in this study include three publicly available datasets: IDD (India Driving Dataset), IRUVD (Indian Road Users Vehicle Dataset), and rsuD20k (Road Scene Understanding Dataset - 20k). While these datasets offer valuable insights into Indian street conditions, several limitations necessitated careful filtering and augmentation to suit the research objectives.

3.3.1 IDD (Indian Driving Dataset)

1. Purpose: A dataset designed to capture diverse traffic scenarios in India, focusing on semantic segmentation tasks. It includes road scenes with varying traffic types, complex scenarios, and unstructured driving conditions unique to Indian roads, primarily from Bengaluru
2. Features:
    - Over 10,000 images with pixel-level annotations.
    - Annotations for challenging road scenes such as crowded traffic, pedestrians, and nonstandard vehicles.

- Comprises 34 classes for semantic segmentation.

3. Challenges:
    - Annotation Issues: Inaccurate bounding boxes were identified during visual inspection, which could lead to erroneous model predictions if not corrected.
    - Repetitive Images: A significant portion of the dataset contains repetitive images of the same scenes with minor variations.
    - Irrelevant Images: Many images included classes irrelevant to the research focus, such as trucks and scooters.
    - Lack of Diversity: The dataset primarily covers street scenes from the city Bengaluru, limiting geographic and cultural diversity.
4. Action Taken: Filtered to retain 745 relevant pedestrian-focused images after manual examination.

3.3.2 IRUVD (Indian Road Users Vehicle Dataset)

1. Purpose: A dataset aimed at understanding and detecting various road users (e.g., pedestrians, cyclists, motorcyclists) and vehicles in Indian conditions, including complex, crowded, and mixed traffic scenarios, primarily from the city Kolkata.
2. Features:
    - Annotated images with bounding boxes for vehicles, pedestrians, and other road users.
    - Focus on capturing unique road user behavior specific to India.
3. Challenges:
    - Static Camera Perspective: Images were captured from static points, resulting in highly repetitive backgrounds and a lack of dynamic perspectives typical of vehicle-mounted cameras.
    - Geographic Limitation: The dataset predominantly represents Kolkata, missing broader diversity in Indian traffic conditions
4. Action Taken: Retained 61 relevant images.

3.3.3 rsuD20k (Road Scene Understanding Dataset - 20k)

1. Purpose: This dataset is used for road scene understanding, object detection, and semantic segmentation, particularly under diverse traffic conditions.
2. Features:
    - Approximately 20,000 images.
    - Covers urban, rural, and semi-urban roads in a Bangladeshi context, which shares cultural and geographical similarities with India.
3. Challenges:
    - Repetitive Images: Only about 10% of the images presented unique scenarios; the rest were repetitive with slight variations..
    - Irrelevant Content: Many images lacked pedestrian instances, which are the primary focus of this study.

- Geographic Limitation: The dataset lacked diversity in its representation of varied urban and rural settings

4. Retained 52 relevant pedestrian-centric images.

3.4 Data Filtering and Final Dataset for Indian Conditions

A thorough manual inspection of all three datasets was conducted to identify and address issues related to labeling accuracy, redundant scenes, and relevance to pedestrian detection. Images and corresponding label files were visually inspected to ensure the correctness of annotations and eliminate redundant or irrelevant images. After this process, a combined dataset was curated with the following composition:

- IDD: 745 images
- IRUVD: 61 images
- rsuD20k: 52 images

3.4.1 Dataset Limitations and Augmentation

Despite the efforts to filter and curate existing datasets, the limited number of images and lack of comprehensive coverage of Indian street conditions presented a significant challenge. To address this, a new dataset was created from scratch, tailored specifically to the objectives of this study.

3.5 Indian Pedestrian Dataset Creation

3.5.1 Video-Based Image Extraction

Given the constraints in capturing live images across India, a systematic approach was adopted to create a new dataset. Publicly available travel videos, selected for their high-quality 4K resolution, were utilized. Videos showcasing diverse locations and people, primarily recorded from two-wheelers or four-wheelers, were chosen to ensure focused visuals of roads and pedestrians. Only videos meeting these criteria were considered to maximize the diversity and quality of the extracted data.

3.5.2 Image Labeling Process

From these videos, 2,173 high-resolution images were manually extracted, including 853 images filtered from the Indian public datasets discussed in the previous section. Each image was then meticulously inspected to annotate all visible pedestrians. Using the LabelImg tool(figure 5.2), bounding boxes were drawn around every pedestrian instance, regardless of size or clarity, ensuring comprehensive coverage.

The label files for each image were saved in .xml format for consistency and future use. This manual process was labor-intensive but critical to the dataset's quality and relevance. By the end of this process, 7,192 pedestrian bounding boxes were annotated.



Figure 3.2: Manual Pedestrain Labeling using LabelImg tool

3.6 Indian Dataset Filtering and Preparation for RQ1 and RQ2

3.6.1 Filtering for RQ1

To ensure consistency with the "reasonable subset" principle, small and unclear bounding boxes were filtered out. Bounding boxes below 50 pixels in height were excluded, reducing noise and maintaining focus on clearly identifiable pedestrians.

3.6.2 Annotations for RQ2

For RQ2, manual annotations were performed to classify bounding boxes into subcategories: male, female, or child. The labeling followed these guidelines:

- Bounding boxes labeled as child were not further classified by gender.
- Adults and older children were labeled as male or female based on visual characteristics.
- Bounding boxes where subcategory determination was not possible were skipped and they have no subcategory

This meticulous process ensured that the dataset was ready for subgroup-based fairness analysis while maintaining high standards of accuracy. Figure 3.3 shows an image from Indian Dataset set created.

3.6.3 Summary Statistics for the Indian Dataset

The final Indian dataset includes the following statistics:

- Total Number of Images: 2,173
- Total Number of Bounding Boxes (Persons): 5,416
- Bounding Boxes by Subcategory:
  - Child: 411
  - Male: 2,577
  - Female: 1,438



Figure 3.3: Labeled Image from Custom Indian Dataset

3.6.4 Publishing the Indian dataset set to publicly available

As a significant outcome of this research, the development of the Indian pedestrian dataset tailored specifically for this study represents a key achievement of this project. Given the limited availability of pedestrian-specific datasets for Indian street conditions, this dataset addresses a critical gap in the field. Furthermore, I plan to expand this dataset to encompass a broader range of scenarios and make it publicly available, contributing to the global research community and fostering advancements in fairness-aware pedestrian detection

4 Modelling

4.1 Model Selection

YOLOX, an advanced general object detection framework, was selected for this research due to its exceptional balance between accuracy, computational efficiency, and adaptability. It employs a fully convolutional neural network architecture capable of real-time detection while maintaining high precision. Key features such as anchor-free prediction, decoupled classification and regression heads, and dynamic label assignment make YOLOX particularly suited for detecting pedestrians in complex and diverse conditions.

YOLOX provides pre-trained models in multiple sizes, including nano, tiny, small, medium, large, and extra-large (YOLOX-n, YOLOX-t, YOLOX-s, YOLOX-m, YOLOX-l, and YOLOX-x, respectively). For this study, we chose the YOLOX-x variant, as it offers an optimal trade-off between accuracy, speed, and resource requirements, making it an excellent choice for analyzing regional biases in pedestrian detection.

Moreover, YOLOX's compatibility with transfer learning allows fine-tuning its pre-trained weights to focus specifically on pedestrian detection tasks. This adaptability is critical for addressing the challenges posed by diverse real-world scenarios, such as crowded environments, varying lighting conditions, and occlusions—features common to both European and Indian street scenes. By leveraging YOLOX, this research ensures robust analysis and meaningful insights into potential biases stemming from dataset composition and regional variations.

4.2 Methodology

The methodology for this study begins with the preparation of two datasets: European street scenes and Indian street scenes. Each dataset is meticulously annotated with ground-truth bounding boxes for pedestrians. After data preparation, the datasets are processed separately through the YOLOX model to perform inference.

During inference, YOLOX predicts bounding boxes for each pedestrian detected in the input images. These predictions are stored as JSON files, with each file containing detailed information about the predicted bounding boxes for individual images. This structured format facilitates systematic analysis and comparison.

Subsequently, the predicted bounding boxes from the JSON files are compared with the ground-truth annotations in the datasets. This comparison is performed using the Intersection over Union (IoU) metric, which quantifies the overlap between predicted and ground-truth bounding boxes. Bounding boxes with an IoU greater than 50% are classified as true positives (TP), while those with an IoU below this threshold are considered false negatives (FN). Figure 4.1 shows an image from Indian Dataset set created.

This analysis is repeated for each image in both datasets, and the resulting TP and FN values are aggregated to calculate key performance metrics, such as the Miss Rate (MR). Furthermore, the comparison process is extended to evaluate model perfor-



Figure 4.1: An inferenced Image - with TP, FN, FP marked

mance across specific demographic subcategories, including male, female, and child. This granular evaluation ensures a detailed understanding of potential biases in pedestrian detection across different regions and demographic groups.

4.3 Evaluation Matrix

In this research, we evaluate pedestrian detection models using standard performance metrics widely adopted in the field. These metrics include Intersection over Union (IoU) and Miss Rate (MR). These measures allow us to quantify the accuracy and effectiveness of the models across diverse datasets and demographic groups.

4.3.1 Intersection over Union (IoU)

The IoU metric is employed to assess whether a detected bounding box sufficiently overlaps with a ground-truth bounding box. It is defined as the ratio of the intersection area to the union area of the detected and ground-truth bounding boxes. Formally IoU is calculated as;

$$IoU = \frac{AreaofOverlap}{AreaofUnion} \tag{4.1}$$

where:

- **Area of Overlap**: The intersection area between the predicted bounding box and the ground truth bounding box.
- **Area of Union**: The combined area of the predicted and ground truth bounding boxes.

In this study, a bounding box is considered successfully detected if the IoU value exceeds 50%. Bounding boxes with an IoU below this threshold are classified as undetected.

4.3.2 Miss Rate (MR)

The Miss Rate quantifies the proportion of ground-truth bounding boxes that remain undetected. It is a key performance indicator in pedestrian detection and is formally expressed as:

$$MR = \frac{FN}{TP + FN} \tag{4.2}$$

where:

- FN: False Negatives – the number of pedestrians missed by the model.
- TP: True Positives – the number of pedestrians correctly detected by the model.
- $TP + FN$: Total number of pedestrians (ground truth instances).

The IoU metric is used as the basis for determining whether a ground-truth bounding box is successfully detected when calculating MR.

4.3.3 Application to Datasets and Subcategories

After inference, the pedestrian detection models produce predicted bounding boxes. These predictions are compared against ground-truth annotations using the **IoU** metric. This process yields the **TP** and **FN** values for the two datasets under consideration: European street scenes and Indian street scenes. The analysis further evaluates the TP and FN values across the sensitive subcategories of male, female, and child.

Using **TP** and **FN** the miss rate **MR** is calculated for the overall data and subcategories within the two datasets.Then using **MR** the detection accuracy is evaluated. This granular approach ensures a comprehensive assessment of fairness and detection accuracy.

4.4 Statistical Analysis

To evaluate the statistical significance of any observed disparities in model performance, particularly concerning fairness, we employ the two-proportion z-test. This test is widely utilized to determine differences between two proportions and is particularly suited for analyzing variations in the miss rate (MR) between different demographic groups.

A result is considered statistically significant if the calculated p-value is below a predefined threshold of 0.05, which is a standard threshold in fairness-related research. For example, when examining whether a significant difference exists between the miss rates for male (MR_{male}) and female (MR_{female}) pedestrians detected by a model, the null hypothesis assumes:

$$H_0 : MR_{male} = MR_{female}$$

If the p-value obtained from the z-test is less than 0.05, we reject the null hypothesis, indicating a statistically significant difference between MR_{male} and MR_{female}.

This statistical methodology is applied consistently across all demographic groups and datasets. By systematically testing for significant differences in miss rates, we aim to identify and quantify potential unfairness in the pedestrian detection models. This approach ensures that our findings are robust, reproducible, and grounded in rigorous statistical analysis.

5 Results and Discussion

5.1 RQ1: Overall Fairness Comparison

RQ1 examines how the pre-trained object detection model **YOLOX** performs in pedestrian detection tasks across datasets from European and Indian street conditions.

Step 1: IoU Evaluation Summary

From the inference results, we obtain the following IoU Evaluation Summary:

- **Indian Dataset:**
  - Total True Positives (TP): 3885
  - Total False Positives (FP): 2580
  - Total False Negatives (FN): 1531
- **European Dataset:**
  - Total True Positives (TP): 8486
  - Total False Positives (FP): 6649
  - Total False Negatives (FN): 596

RQ1: European Dataset Metrics Summary

Metric	Value	Observation
Precision	0.56	Moderate precision; many false positives dilute accuracy.
Recall	0.93	High recall indicates most true pedestrians are detected.
F1-Score	0.70	F1-Score suggests balanced performance, weighted by precision and recall.
Miss Rate	0.07	Low miss rate reflects good pedestrian detection coverage.
False Positives	6649	High false positives indicate over-detection and misclassification.
False Negatives	596	Low false negatives indicate that most ground truth pedestrians are detected.
Accuracy	0.54	Moderate accuracy shows room for reducing false positives.

RQ1: Indian Dataset Metrics Summary

Metric	Value	Observation
Precision	0.6009	Moderate precision; false positives reduce the reliability of detections.
Recall	0.7173	Good recall; most true pedestrians are successfully detected.
F1-Score	0.6540	Balanced performance indicated by the F1-Score, considering precision and recall.
Miss Rate	0.2827	Moderate miss rate suggests some pedestrians are missed in the detection process.
False Positives	2580	False positives indicate over-detection or misclassification of objects.
False Negatives	1531	False negatives highlight the missed detection of true pedestrians.
Accuracy	0.5385	Moderate accuracy shows potential for improvement by reducing false positives and negatives.

Figure 5.1: Some relevant Evaluation Matrics for India and Europe Dataset

Step 2: Evaluation Metrics and Miss Rate

Using the IoU Evaluation Summary, we calculate the following metrics for each dataset:

- **Miss Rate (MR):**

$$MR = \frac{FN}{TP + FN} \tag{5.1}$$

For Indian Dataset:

$$MR_{India} = \frac{FN_{India}}{TP_{India} + FN_{India}} = \frac{1531}{3885 + 1531} \approx 0.2827$$

For European Dataset:

$$MR_{Europe} = \frac{FN_{Europe}}{TP_{Europe} + FN_{Europe}} = \frac{596}{8486 + 596} \approx 0.0656$$

Step 3: Compare Miss Rates

The Miss Rate for the Indian dataset ($MR_{India} = 0.2827$) is significantly higher than that for the European dataset ($MR_{Europe} = 0.0656$). This indicates that the pedestrian detection model misses more pedestrians in the Indian dataset compared to the European dataset, suggesting a potential fairness issue.

Step 4: Perform Statistical Analysis

To determine whether the observed difference in Miss Rates is statistically significant, we perform a two-proportion z-test. The input values for the test are as follows:

- $p_{India} = MR_{India} = 0.2827$
- $p_{Europe} = MR_{Europe} = 0.0656$
- $n_{India} = 5416$
- $n_{Europe} = 9082$

The z-score is calculated as:

$$Z = \frac{p_{India} - p_{Europe}}{\sqrt{p_{pooled} \cdot (1 - p_{pooled}) \cdot \left(\frac{1}{n_{India}} + \frac{1}{n_{Europe}}\right)}}$$

Where the pooled proportion (p_{pooled}) is:

$$p_{pooled} = \frac{FN_{India} + FN_{Europe}}{n_{India} + n_{Europe}} = \frac{1531 + 596}{5416 + 9082} \approx 0.1467$$

Substituting the values:

$$Z \approx \frac{0.2827 - 0.0656}{0.005} \approx 43.42$$

The p-value corresponding to $Z = 43.42$ is effectively $p \approx 0.0$ (extremely small).

Step 5: Hypothesis Testing

Hypotheses:

$$H_0 : MR_{India} = MR_{Europe}$$

$$H_1 : MR_{India} \neq MR_{Europe}$$

Decision Rule: If $p < 0.05$, reject H_0; otherwise, fail to reject H_0.

Result: Since $p \approx 0.0 < 0.05$, the null hypothesis is rejected. This confirms that the observed difference in Miss Rates between the Indian and European datasets is statistically significant.

5.1.1 Answer to RQ1

Ans: The pre-trained object detection model YOLOX exhibits significant unfairness in pedestrian detection tasks when applied to datasets from Indian and European street conditions. The Miss Rate for the Indian dataset ($MR_{India} = 0.2827$) is significantly higher than that for the European dataset ($MR_{Europe} = 0.0656$), indicating that the model performs worse in Indian street conditions. This difference is statistically significant ($p \approx 0.0$), suggesting that the observed disparity is unlikely to be due to random chance.

5.2 RQ2: Category wise fairness Comparison

To assess the fairness of pre-trained pedestrian detection models across common sensitive groups (**Male**, **Female**, and **Children**), we conducted both **within-dataset group comparisons** and **across-dataset group comparisons**. These analyses aimed to identify potential gender- and age-based fairness issues and regional biases in detection performance. The calculations are similar to what is explained for RQ1. We skip the calculation part here.

5.2.1 Within-Dataset Group Comparisons

Case 1: Indian Male vs. Indian Female
Z = -3.2387, p = 0.0012
The result is statistically significant ($p < 0.05$), indicating a difference in Miss Rates between Indian Male and Indian Female groups.
Inference: The model exhibits **gender-based unfairness** in the Indian dataset, with a significant disparity in detection performance between Male and Female pedestrians.

Case 2: Indian Adult vs. Indian Child
Z = -0.6566, p = 0.5114
The result is not statistically significant ($p \geq 0.05$), indicating no significant difference in Miss Rates between Indian Adults (combined Male and Female) and Indian Children.
Inference: The model's performance does not exhibit **age-based unfairness** in the Indian dataset. However, fairness concerns persist for both Adults and Children.

Case 3: European Male vs. European Female
Z = -0.0613, p = 0.9511
The result is not statistically significant ($p \geq 0.05$), indicating no significant difference in Miss Rates between European Male and Female pedestrians.
Inference: The model does not exhibit **gender-based unfairness** in the European dataset.

Case 4: European Adult vs. European Child
Z = -22.5707, p = 0.0000
The result is statistically significant ($p < 0.05$), indicating a significant difference in Miss Rates between European Adults (combined Male and Female) and European Children.
Inference: The model exhibits **age-based unfairness** in the European dataset, with poorer detection performance for Children compared to Adults.

Our findings specific to European conditions covered in Case 3 and Case 4 are in parallel with a similar study done by Li et al,[12] where they found significance proportion of undetected children compared to adults.

5.2.2 Across-Dataset Group Comparisons

Case 5: Indian Male vs. European Male
Z = 19.5092, p = 0.0000
The result is statistically significant ($p < 0.05$), indicating a significant difference in Miss Rates for Male pedestrians between the Indian and European datasets.
Inference: The model performs differently for Male pedestrians across regions, with **better performance in Europe**.

Case 6: Indian Female vs. European Female
Z = 19.2861, p = 0.0000
The result is statistically significant ($p < 0.05$), indicating a significant difference in Miss Rates for Female pedestrians between the Indian and European datasets.
Inference: The model exhibits **regional biases** for Female pedestrians, with **better performance in Europe**.

Case 7: Indian Child vs. European Child
Z = -3.4516, p = 0.0006
The result is statistically significant ($p < 0.05$), indicating a significant difference in Miss Rates for Children between the Indian and European datasets.
Inference: The model exhibits **regional biases** for Children, with **better performance in Europe**.

5.2.3 Answer to RQ2

To what extent do pre-trained pedestrian detectors exhibit unfairness concerning common sensitive groups (**Male**, **Female**, **Children**)?

Within-Dataset Fairness

- The model exhibits **gender-based unfairness** in the Indian dataset, with a significant disparity in detection performance between Male and Female pedestrians.
- The model exhibits **age-based unfairness** in the European dataset, with significantly poorer performance for Children compared to Adults.
- No significant unfairness is observed for age groups (Adults vs. Children) in the Indian dataset or for gender groups (Male vs. Female) in the European dataset.

Across-Dataset Fairness

- Significant differences in Miss Rates are observed for all three sensitive groups (**Male**, **Female**, and **Children**) between the Indian and European datasets, indicating **regional biases** in detection performance.
- These disparities suggest that the model performs **better in European conditions** compared to Indian conditions for all groups.

A summary table for the discussion is given in figure 5.2

RQ2: Category-wise Fairness Analysis

Comparison	Z-Score	P-Value	Significance	Observation
Indian Male vs. Indian Female	-3.2387	0.0012	Significant	Significant gender-based fairness issue in India.
Indian Adult vs. Indian Child	-0.6566	0.5114	Not Significant	No significant age-based fairness issue in India, but fairness concerns persist for children.
European Male vs. European Female	-0.0613	0.9511	Not Significant	No significant gender-based fairness issue in Europe.
European Adult vs. European Child	-22.5707	0.0	Significant	Significant age-based fairness issue in Europe, with children being disproportionately affected.
Indian Male vs. European Male	19.5092	0.0	Significant	Significant regional fairness issue for males, with better performance in Europe.
Indian Female vs. European Female	19.2861	0.0	Significant	Significant regional fairness issue for females, with better performance in Europe.
Indian Child vs. European Child	-3.4516	0.0006	Significant	Significant regional fairness issue for children, with better performance in Europe.

Figure 5.2: RQ2 Summary

6 Conclusion and Future work

6.1 Conclusion

This study reveals significant fairness issues in pedestrian detection performance for Indian street conditions when using the YOLOX model. The higher Miss Rates observed in the Indian dataset compared to the European dataset ($MR_{India} = 0.2827$ vs. $MR_{Europe} = 0.0656$) confirm a statistically significant disparity, highlighting regional biases in the model's detection capabilities. Additionally, the category-wise fairness analysis indicates that the model exhibits fairness issues across all subgroups in the Indian dataset, with significant gender-based unfairness between Male and Female pedestrians, and persistent concerns for Children. In contrast, for the European dataset, the model primarily shows age-based unfairness, with poorer detection performance for Children compared to Adults. These findings underscore the critical need to address demographic and subgroup-based biases in pedestrian detection systems.

These findings emphasize the critical need to address fairness across demographic and sensitive subgroups in pedestrian detection tasks. While this study focuses on the YOLOX model, the insights and methodologies presented can be instrumental in designing and evaluating other detection models, particularly for autonomous driving applications. Ensuring fairness in such systems is essential to avoid exacerbating real-world safety risks for underrepresented or underserved groups.

6.2 Self-Appraisal and Future Directions

6.2.1 Self-Appraisal

This research began with the ambitious goal of analyzing fairness issues in pedestrian detection across different demographic regions, identifying root causes, and proposing mitigating measures. The initial plan was to evaluate multiple pedestrian detection models, including general object detection models and pedestrian-specific models, to achieve a broader understanding of fairness in detection systems. However, due to time constraints, the scope of the study evolved into a focused fairness analysis using a single general object detection model, YOLOX. While YOLOX demonstrated impressive performance, it lacks the domain-specific optimizations of pedestrian-specific models, such as ALFNet, CSP, and PRNet, which are tailored to handle varying pedestrian poses, sizes, and occlusion levels.

A significant challenge encountered during the research was the lack of appropriate datasets for Indian street conditions. Substantial time and effort were dedicated to data preparation, including manually gathering and labeling data to create a tailored dataset. While this dataset served as a valuable foundation, the limited scope of models analyzed constrains the generalizability of the findings to other pedestrian detection frameworks. Nevertheless, this study provides crucial insights into potential biases and fairness issues that may arise when using a general object detection framework like YOLOX in diverse

regional contexts. By doing so, it highlights the importance of fairness-aware design in pedestrian detection systems, particularly for autonomous driving applications.

6.2.2 Future Works

The unaccomplished aspects of the original proposal will now shape the core of my future work. My future goals include:

- **Expanding the Dataset**: Developing a more extensive and diverse dataset for Indian conditions, covering a broader range of scenarios such as varying weather, lighting, and traffic densities. Once complete, I aim to make this dataset publicly available to support fairness research in pedestrian detection and autonomous driving.
- **Broader Model Evaluation**: Evaluating fairness across a variety of models, including other general object detection models like RetinaNet and Faster R-CNN, and pedestrian-specific models such as ALFNet and CSP. This will allow for a more comprehensive understanding of fairness across different architectures and training methodologies.
- **Exploring the Root Causes of Fairness Issues**: Investigating the reasons behind fairness issues by analyzing model decisions in cases of *True Positives*, *False Positives*, and *False Negatives*. Using explainable AI techniques, I aim to identify specific features in pedestrians (e.g., clothing, size, or occlusion) that influence detection outcomes. For example, I aspire to pinpoint why a False Negative occurred and attribute it to features in the pedestrian or environmental conditions.

This research journey has instilled a deep passion for improving fairness in pedestrian detection and ensuring that such systems are equitable across all demographic groups and conditions. While this thesis encountered genuine limitations in scope, the findings underscore the importance of fairness-aware evaluation in autonomous systems.

Bibliography

[1] Joshua Garcia, Yang Feng, Junjie Shen, Sumaya Almanee, Yuan Xia, and Qi Alfred Chen. A comprehensive study of autonomous vehicle bugs. In *Proceedings of the 42nd International Conference on Software Engineering, ICSE 2020*, pages 385–396, 2020.

[2] The New York Times. Self-driving uber car kills pedestrian in arizona, where robots roam, 2018. https://www.nytimes.com/2018/03/19/technology/uber-driverless-fatality.html.

[3] Yuchi Tian, Kexin Pei, Suman Jana, and Baishakhi Ray. Deeptest: Automated testing of deep-neural-network-driven autonomous cars. In *Proceedings of the 40th International Conference on Software Engineering (ICSE 2018)*, pages 303–314, New York, NY, USA, 2018. ACM.

[4] Mengshi Zhang, Yuqun Zhang, Lingming Zhang, Cong Liu, and Sarfraz Khurshid. Deeproad: GAN-based metamorphic testing and input validation framework for autonomous driving systems. In *Proceedings of the 33rd ACM/IEEE International Conference on Automated Software Engineering (ASE 2018)*, pages 132–142, New York, NY, USA, 2018. ACM.

[5] Husheng Zhou, Wei Li, Zelun Kong, Junfeng Guo, Yuqun Zhang, Bei Yu, Lingming Zhang, and Cong Liu. Deepbillboard: Systematic physical-world testing of autonomous driving systems. In *Proceedings of the ACM/IEEE 42nd International Conference on Software Engineering (ICSE 2020)*, pages 347–358, New York, NY, USA, 2020. ACM.

[6] Zhenpeng Chen, Jie M. Zhang, Max Hort, Mark Harman, and Federica Sarro. Fairness testing: A comprehensive survey and analysis of trends. *ACM Transactions on Software Engineering and Methodology*, pages 137:1–137:59, 2024.

[7] Andrew D. Selbst, Danah Boyd, Sorelle A. Friedler, Suresh Venkatasubramanian, and Janet Vertesi. Fairness and abstraction in sociotechnical systems. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* 2019)*, pages 59–68, 2019.

[8] Markus Braun, Sebastian Krebs, Fabian Flohr, and Dariu M. Gavrila. EuroCity Persons: A Novel Benchmark for Person Detection in Traffic Scenes. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 41(8):1844–1861.

[9] Martim Brandao. Age and gender bias in pedestrian detection algorithms. *https://arxiv.org/abs/1906.10490*.

[10] Benjamin Wilson, Judy Hoffman, and Jamie Morgenstern. Predictive inequity in object detection. *arXiv preprint arXiv:1902.11097*, 2019.

[11] Shunsuke Kogure, Kai Watabe, Ryosuke Yamada, Yoshimitsu Aoki, Akio Nakamura, and Hirokatsu Kataoka. Age should not matter: Towards more accurate pedestrian detection via self-training.

[12] Xinyue Li, Zhenpeng Chen, Jie M. Zhang, Federica Sarro, Ying Zhang, and Xuanzhe Liu. Bias behind the wheel: Fairness testing of autonomous driving systems. *ACM Transactions on Software Engineering and Methodology*, 33(6), 2024.

[13] S. Nisha, Aishwarya M., and Manohar N. Detection of vehicle and pedestrian in indian challenging scenarios. In *Proceedings of the 2024 International Conference on Vehicular Technology and Transportation Systems (ICVTTS)*, 2024.